UNIT – II

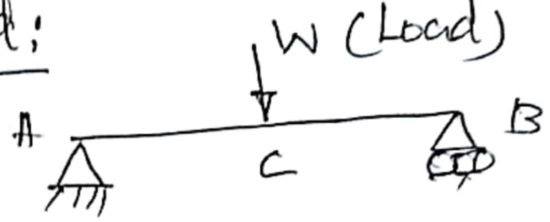
part A (Two Marks)

1. what are the types of loads with sketches (May 2016)

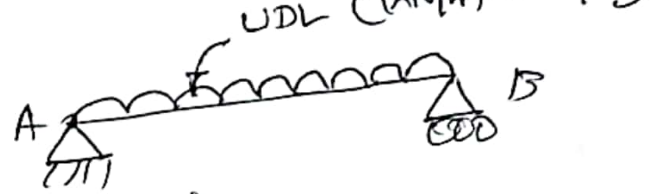
Types of loads:

1. point load 2. Uniformly distributed load
3. Uniformly varying load.

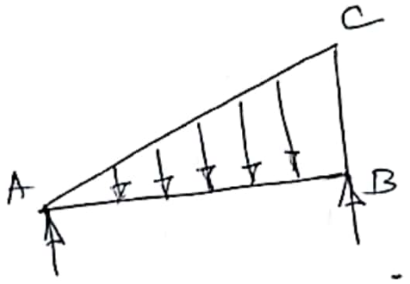
1. point load:



2. Uniformly distributed load (UDL):



3. Uniformly varying load:



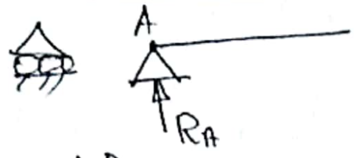
2. Write down the types of support in two dimensions with sketches (May 2016), (May 2017) (Dec. 2018)

Types of supports:

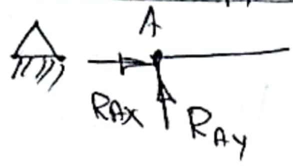
1. Simple support
2. Roller support
3. Hinged support
4. Fixed support

1. Simple support:

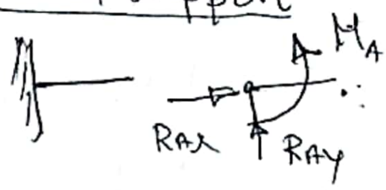
2. Roller support



3. Hinged support



4. Fixed support



3. Define angle of repose and angle of friction (May 2016)

Angle of repose:

The angle between an inclined plane and horizontal plane when a body placed on it is just on the point of moving down is called angle of repose.

Angle of friction:

It is defined as the maximum angle between normal reaction and the resultant of the limiting friction.

4. What are the different types of methods to analysis the members of a structure? (May 2016)

1. Method of Joint
2. Method of Section
3. Graphical method,

5. Define the coefficient of static friction (Nov. 2016)

Co-efficient of static friction (μ_s)

It is defined as the ratio of the limiting friction (F_s) to the normal reaction (R)

$$\mu_s = \frac{F_s}{R}$$

6. Define limiting force of friction (Dec. 2017)

If the frictional force 'F' shall go on increasing to balance the increasing applied force 'P', the body shall remain at rest. Then, there comes a limit beyond which the frictional force 'F' cannot increase and the body begins to move. The frictional force 'F' at this instant is called as limiting friction.

7. State the coloumb's law of dry friction (May 2019) (May 2017, May 2018)

i) The force of friction always acts in a opposite direction to that in which the body tends to move.

ii) The magnitude of the force of friction is always equal to the applied force.

iii) The force of friction depends upon the roughness of the surface.

iv) The force of friction is independent of the area of contact between the bodies.

8. Define impending motion (Dec. 2018)

The state of motion of a body which is just about to move or slide is called impending motion.

9. Differntiate between a space diagram and a free body diagram (May 2019).

Free body diagram:

It is a diagram which shows all the Kind of forces such as action, reaction force weight etc. by the eleminating the supports

space diagram:

A drawing of a structure that indicates it's form as well as means of it's support and loading conditions.

10. Defcine degree of freedom (May 2018, Dec 2017, Jan. 2016)

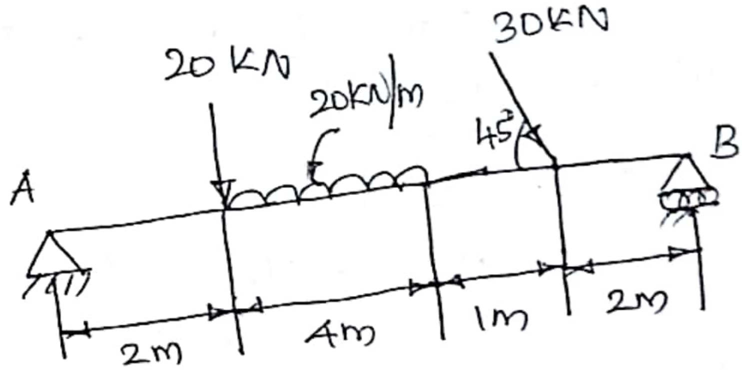
The direction that a joint can move or rotates is called degree of freedom.

UNIT – II

part – B

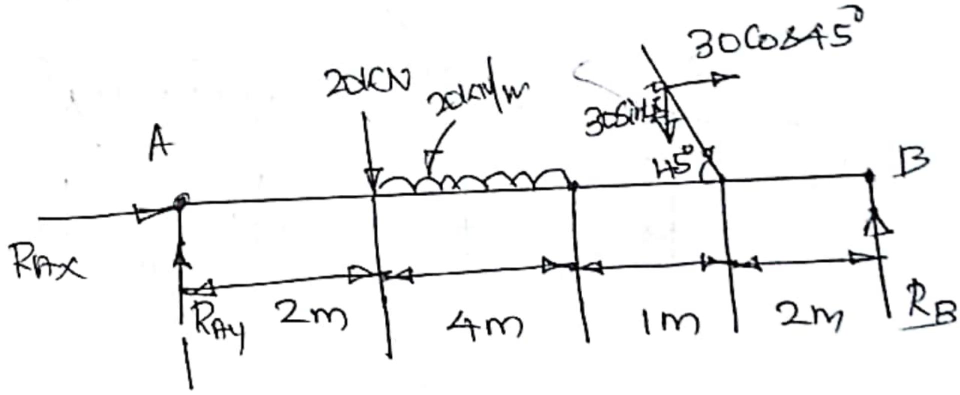
Support Reactions

1. Find the reactions at support A and B of the loaded beam shown in fig. (Jan 2016)



Solutions:

FBD



$\Sigma F_x = 0$

$R_{Ax} + 30\cos 45^\circ = 0$

$R_{Ax} = -30\cos 45^\circ$

$= -21.21\ KN$

$\therefore = 21.21\ KN\ (\leftarrow)$

$\Sigma F_y = 0$

$R_{Ay} + R_B - 20 - 30 \times 4 - 30\sin 45^\circ = 0$

$R_{Ay} + R_B = 140 + 30\sin 45^\circ$

$R_{Ay} + R_B = 161.21$ ——— (1)

$\Sigma M_A = 0$

$R_B \times 9 - (20\times2) - (120\times4) - (30\sin45^\circ \times 7) = 0$

$R_B \times 9 - 40 - 480 - 148.49 = 0$

$R_B \times 9 = 668.49$

$R_B = \frac{668.49}{9} = \underline{74.276\ KN}$

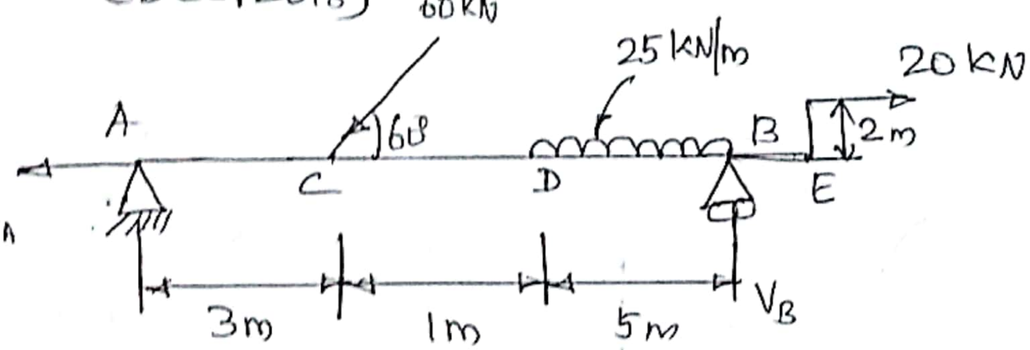
Sub. in (1)

$R_{Ay} + 74.276 = 161.21$

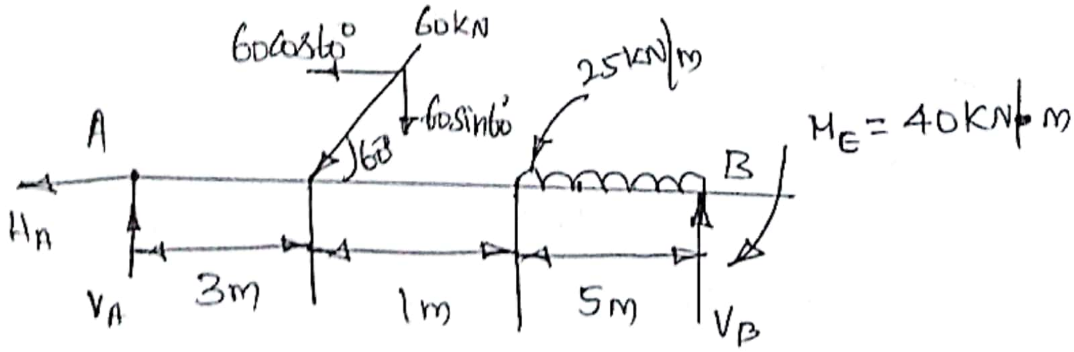
$R_{Ay} = 161.21 - 74.276$

$= \underline{86.934\ KN}$

2. Determine the reactions of the beam as shown in fig. (Dec, 2018)



FBD



$\Sigma F_x = 0$

$-H_A - 60\cos60^\circ + 20 = 0$

$-H_A - 30 + 20 = 0$

$H_A = -10\ KN$

$\therefore H_A = 10\ KN\ (\leftarrow)$

Scanned with CamScanner

$\Sigma F_y = 0$

$V_A + V_B - 60\sin 60^\circ - 25\times 5 = 0$

$V_A + V_B = 60\sin 60^\circ + 25\times 5$

$= 51.96 + 125$

$= 176.96$ —— (1)

$\Sigma M_A = 0$

$V_B \times 9 - (60\sin 60^\circ \times 3) - (125\times 6.5) - 40 = 0$

$V_B \times 9 - 155.88 - 812.5 - 40 = 0$

$V_B \times 9 = 1008.38$

$V_B = \frac{1008.38}{9} = 112.04$ KN

sub. in (1),

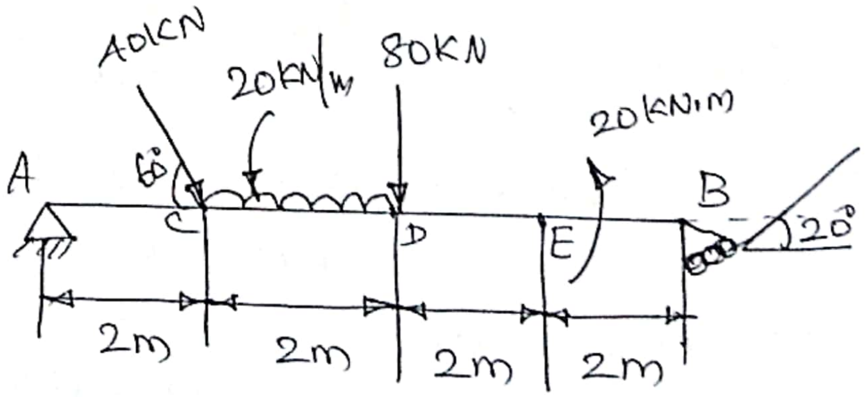
$V_A + V_B = 176.96$

$V_A + 112.04 = 176.96$

$V_A = 176.96 - 112.04$

$= 64.92$ KN

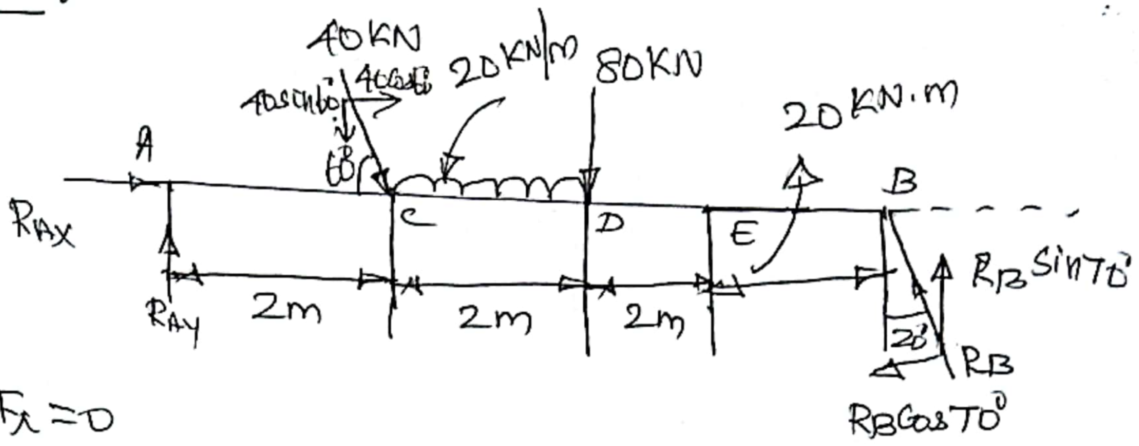
3. Find the reactions at the supports A and B of the beam shown in fig. (DEC.2017)



Scanned with CamScanner

Solution:

FBD



$$\Sigma F_x = 0$$

$$R_{AX} - R_B \cos 70^\circ + 40\cos 60^\circ = 0$$

$$R_{AX} = R_B \cos 70^\circ - 40\cos 60^\circ$$

$$R_{AX} = 0.342 R_B - 20 \quad \text{——— (1)}$$

$$\Sigma F_y = 0$$

$$R_{AY} + R_B \sin 70^\circ - 40\sin 60^\circ - 40 - 80 = 0$$

$$R_{AY} + 0.939 R_B - 34.64 - 40 - 80 = 0$$

$$R_{AY} + 0.939 R_B = 154.64$$

$$\Sigma M_A = 0$$

$$(R_B \sin 70^\circ \times 8) - (40\sin 60^\circ \times 2) - (40 \times 3) - (80 \times 4) + 20 = 0$$

$$7.51 R_B - 69.28 - 120 - 320 + 20 = 0$$

$$7.51 R_B = 489.28$$

$$R_B = \frac{489.28}{7.51} = \underline{65.15 \text{ KN}}$$

Sub. in (1),

$$R_{AX} = 65.15 \times 0.342 - 20$$

$$= 22.28 - 20 = \underline{2.28 \text{ KN}}$$

Sub. R_B in (2),

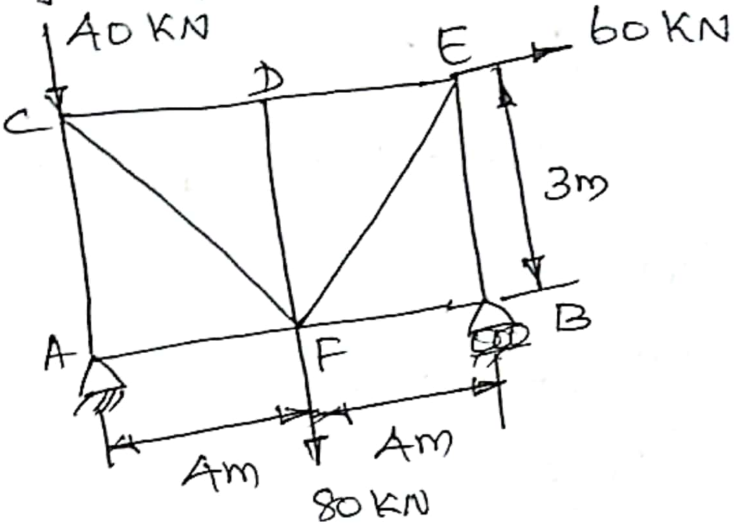
$$R_{AY} + R_B\, 0.939 = 154.64$$

$$R_{AY} + 65.15 \times 0.939 = 154.64$$

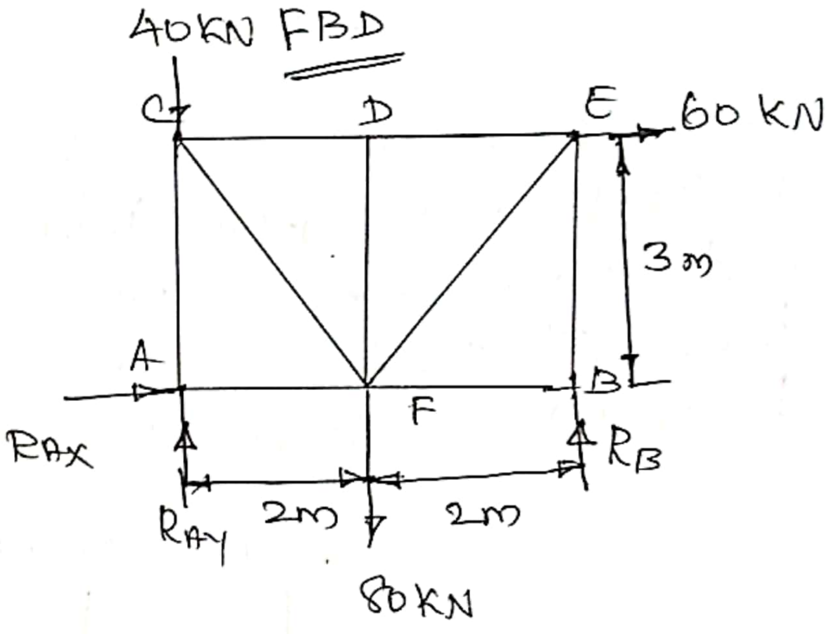
$$R_{AY} = \underline{93.46 \text{ KN}}$$

Truss Analysis

1. Find the support Reactions of truss as shown in fig. (May 2016)



Solution:



$\Sigma F_x = 0$

$R_{Ax} + 60 = 0$

$R_{Ax} = -60$ KN

$\therefore = 60$ KN ($\leftarrow$)

$\Sigma F_y = 0$

$R_{Ay} + R_B - 80 - 40 = 0$

$R_{Ay} + R_B = 120$ ——— (1)

$\Sigma M_A = 0$

$(R_B \times 8) - (80 \times 4) - (60 \times 3) = 0$

$R_B \times 8 - 320 - 180 = 0$

$R_B \times 8 = 500$

$R_B = \frac{500}{8} = \underline{\underline{62.5 \text{ KN}}}$

Sub. in ①,

$R_{AY} + R_B = 120$

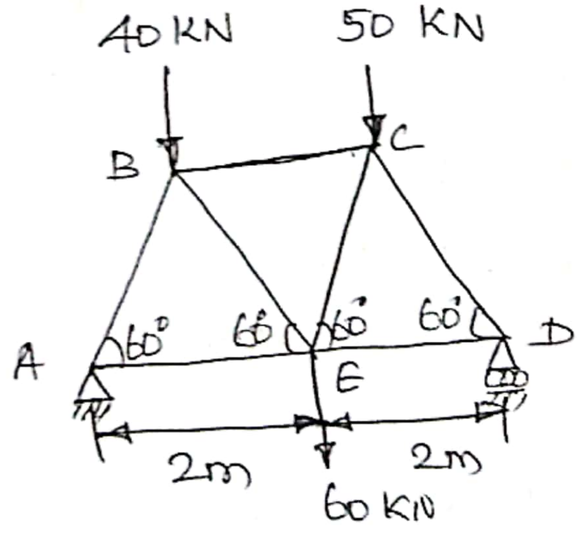
$R_{AY} + 62.5 = 120$

$R_{AY} = 120 - 62.5$

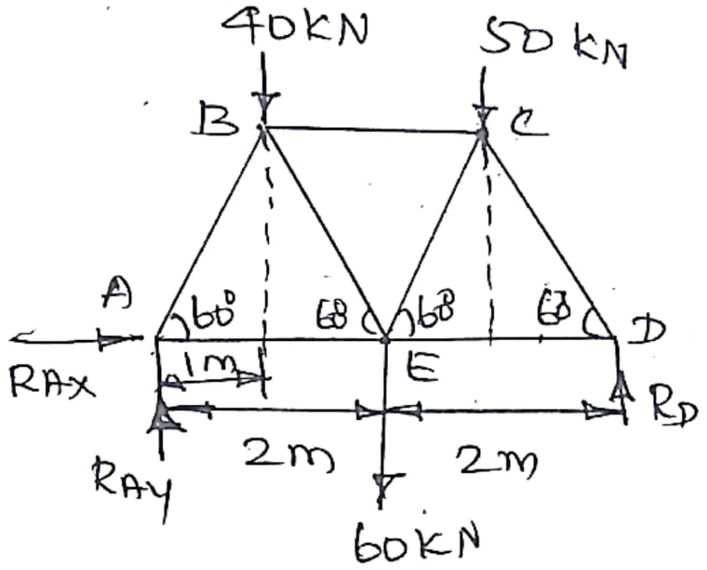
$= \underline{\underline{57.5 \text{ KN}}}$

2. Determine the forces in all the members of the truss in fig. and indicate the magnitude and nature of forces on the diagram of the truss. All inclined members are at 60° to the horizontal and length of each member is 2m

(NOV. 2016)



FBD



$\Sigma F_x = 0$

$R_{AX} = 0$

$\Sigma F_y = 0$

$R_{AY} + R_D - 60 - 40 - 50 = 0$

$R_{AY} + R_D = 150$ —— (1)

$\Sigma M_A = 0$

$(R_D \times 4) - (60 \times 2) - (40 \times 1) - (50 \times 3) = 0$

$R_D \times 4 = 310$

$R_D = \frac{310}{4} = 77.5$ KN

Sub. in (1),

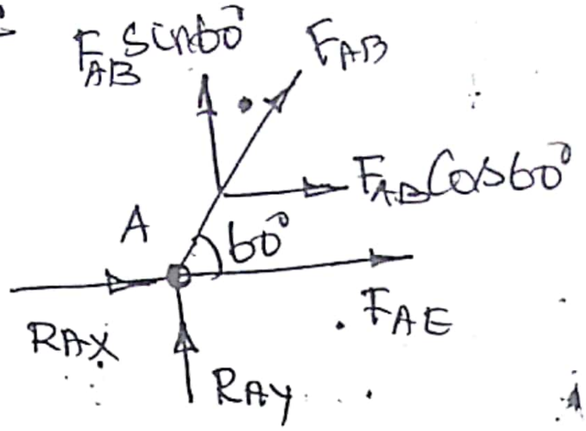
$R_{AY} + R_D = 150.$

$R_{AY} + 77.5 = 150$

$R_{AY} = 150 - 77.5$

$= 72.5$ KN

Joint A:



$\Sigma F_x = 0$

$R_{Ax} + F_{AB}\cos 60^\circ + F_{AE} = 0$

$0 + F_{AB}\cos 60^\circ + F_{AE} = 0$

$F_{AE} + F_{AB}\cos 60^\circ = 0$ ——— (2)

$\Sigma F_y = 0$

$R_{Ay} + F_{AB}\sin 60^\circ = 0$

$72.5 + F_{AB}\sin 60^\circ = 0$

$F_{AB} = \dfrac{-72.5}{\sin 60^\circ} = -83.718$ KN

$= 83.718$ KN (C)

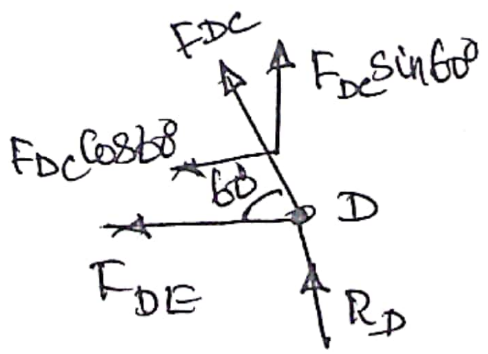
sub. in (2),

$F_{AE} + F_{AB}\cos 60^\circ = 0$

$F_{AE} - 83.718\cos 60^\circ = 0$

$F_{AE} = 41.859$ KN (T)

Joint D:



$\Sigma F_x = 0$

$-F_{DE} - F_{DC}\cos 60^\circ = 0$ —— (3)

$\Sigma F_y = 0$

$R_D + F_{DC}\sin 60^\circ = 0$

$72.5 + F_{DC}\,0.866 = 0$

$F_{DC} = \dfrac{-72.5}{0.866} = -83.718$

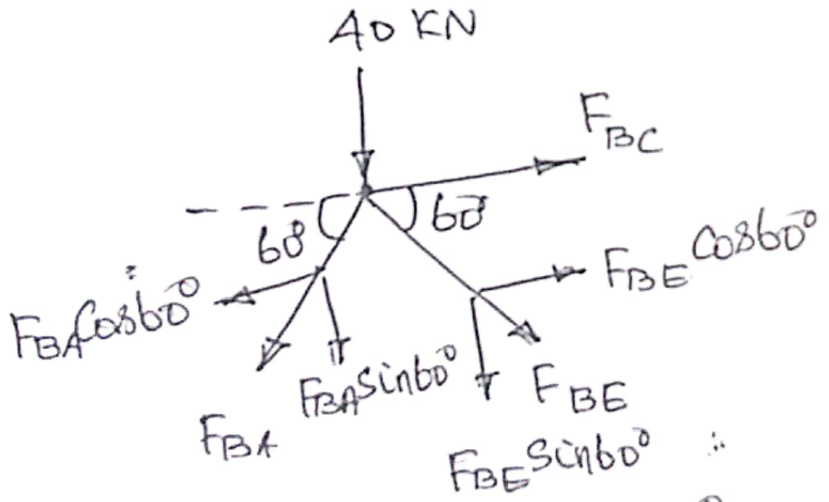
$= 83.718\ \text{kN}\ (C)$

sub. in (3)

$-F_{DE} - (-83.718)\cos 60^\circ = 0$

$F_{DE} = 41.859\ \text{kN}\ (T)$

Joint B:



$\Sigma F_x = 0$

$F_{BC} + F_{BE}\cos 60^\circ - F_{BA}\cos 60^\circ = 0$

$F_{BC} + F_{BE}\cos 60^\circ - (-83.718)\cos 60^\circ = 0$

$F_{BC} + F_{BE}\cos 60^\circ + 83.718\cos 60^\circ = 0$

$F_{BC} + F_{BE}\cos 60^\circ = -41.859$ —— (4)

$F_{BC} + F_{BE}\cos 60^\circ$

$\Sigma F_y = 0$

$-F_{BA}\sin 60^\circ - F_{BE}\sin 60^\circ - 40 = 0$

$-(-83.718)\sin 60^\circ - F_{BE}\sin 60^\circ = 40$

$72.5 - F_{BE}\sin 60^\circ = 40$

$-F_{BE}\sin 60^\circ = 40 - 72.5$

$+F_{BE}\sin 60^\circ = 32.659$

$F_{BE} = \dfrac{32.659}{0.866} = 37.659$

$= 37.659 \text{ KN (T)}$

Sub. in ④,

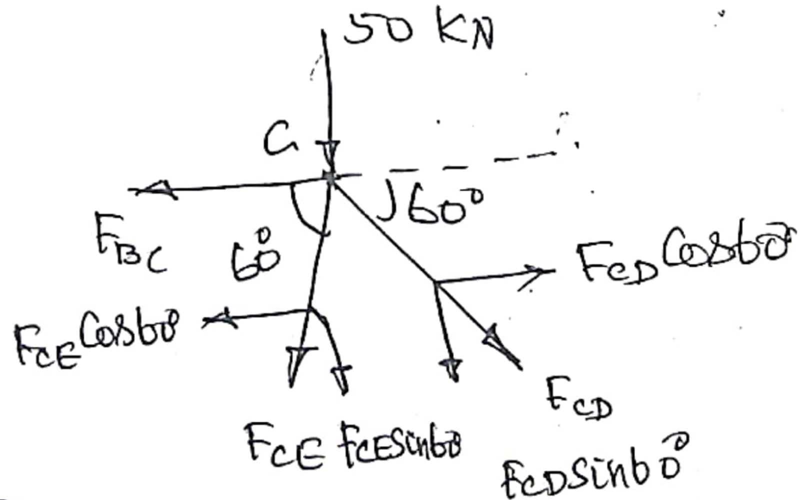
$F_{BC} + F_{BE}\cos 60^\circ = -41.859$

$F_{BC} + 37.659\cos 60^\circ = -41.859$

$F_{BC} = -60.688$

$F_{BC} = 60.688 \text{ KN (C)}$

Joint C:



$\Sigma F_x = 0$

$-F_{BC} - F_{CE}\cos 60^\circ + F_{CD}\cos 60^\circ = 0$

$-(-60.688) - F_{CE}\cos 60^\circ + (-83.718)\cos 60^\circ = 0$

$60.688 - F_{CE}\cos 60^\circ - 41.859 = 0$

$F_{CE} = 37.658 \text{ kN (T)}$

Table:

S.No	Member	Magnitude of force (KN)	Nature of force
1.	AB	83.718	C
2.	AE	41.859	T
3.	BC	60.688	C
4.	BE	37.659	T
5.	CD	83.718	C
6.	CE	37.658	T
7.	DE	41.859	T

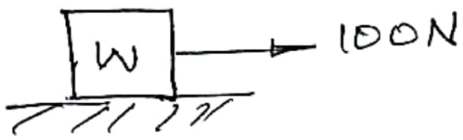
C - Compressive Force

T - Tensile Force

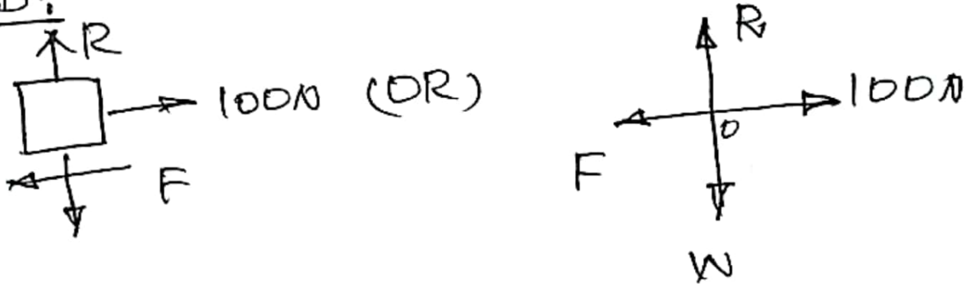
1. A body lying on a horizontal plane is moved slowly along the plane when a force of 100N is applied to the body. Also a force of 90N is applied to the body at an angle of 30° to the horizontal, moves the body slowly along the same plane. Determine the weight of the body and coefficient of friction.

Solution:

<u>case 1:</u>



<u>FBD:</u>



$\Sigma F_x = 0$

$-F + 100 = 0$

$F = 100\,N$

$\Sigma F_y = 0$

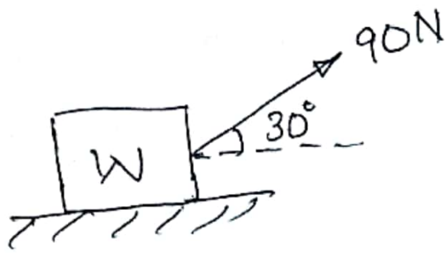
$R - W = 0$

$R = W$

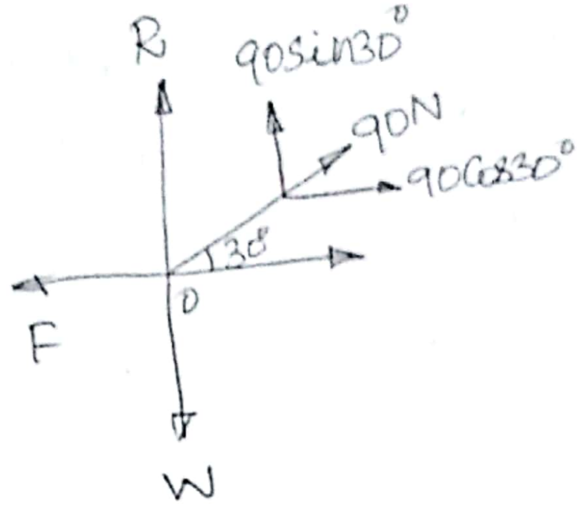
$F = \mu R$

$100 = \mu W$ —— (1)

<u>case 2:</u>



FBD:



$\Sigma F_x = 0$

$-F + 90\cos30° = 0$

$F = 90\cos30°$

$= 78N$

$\Sigma F_y = 0$

$R - W + 90\sin30° = 0$

$R - W + 45 = 0$

$R = -45 + W$

$R = W - 45$ —— (2)

$F = \mu R$ —— (2)

Sub. R in (2)

$78 = \mu R$

$78 = \mu(W - 45)$ —— (3)

$\frac{(1)}{(3)} \Rightarrow \frac{100}{78} = \frac{\cancel{\mu} W}{\cancel{\mu}(W-45)}$

$1.28 = \frac{W}{W-45}$

$1.28(W-45) = W$

$1.28W - W = 57.69$

$W(1.28-1) = 57.69$

$W(0.28) = 57.69$

$W = \frac{57.69}{0.28} = \underline{\underline{206\,N}}$

Sub in (1),

$100 = \mu W$

$= 206\mu$

$\mu = \frac{100}{206} = \underline{\underline{0.48}}$

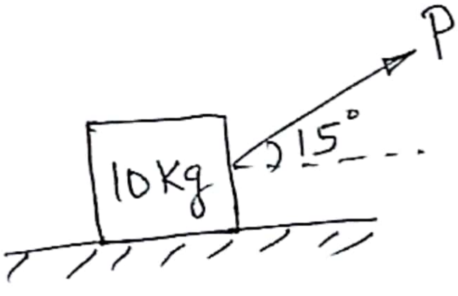
i) Weight of the body, $W = \underline{\underline{206\,N}}$

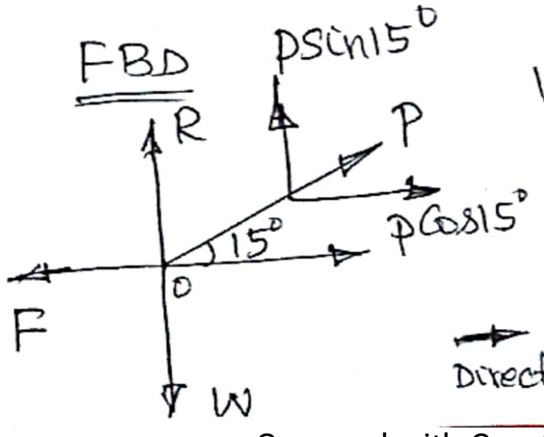
ii) Co-efficient of friction $\mu = \underline{\underline{0.48}}$

2. A block of weight 10 kg is kept on a horizontal plane. Find the force required to cause motion, if the applied force is 15° with the horizontal plane. Take the co-efficient of friction as 0.25 (May 2018)

Solution:

$\mu = 0.25$





$W = mg$

$= 10 \times 9.81$

$= \underline{\underline{98.1\,N}}$

Direction of motion

$\Sigma F_x = 0$

$-F + P\cos 15^\circ = 0$ | $F = \mu R$

$F = P\cos 15^\circ$

$\mu R = 0.965P$ ——— (1)

$\Sigma F_y = 0$

$R - 98.1 + P\sin 15^\circ = 0$

$R = 98.1 - 0.258P$

Sub in (1)

$\mu R = 0.965P$

$0.25R = 0.965P$

$0.25(98.1 - 0.258P) = 0.965P$

$24.52 - 0.064P = 0.965P$

$0.064P + 0.965P = 24.52$

$P(0.064 + 0.965) = 24.52$

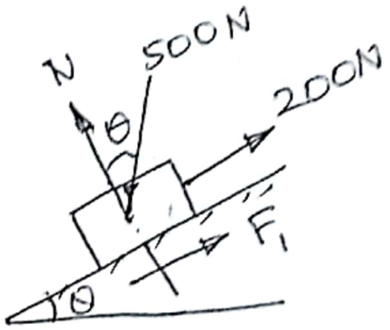
$P(1.029) = 24.52$

$P = \frac{24.52}{1.029}$

$= \underline{\underline{23.82\ N}}$

3. A weight 500N just starts moving down a rough inclined plane supported by a force of 200N acting parallel to the plane and it is at the point of moving up the plane when pulled by a force of 300N parallel to the plane. Find the inclination of the plane and the co-efficient of friction between the inclined plane and the weight. (NOV.2016)

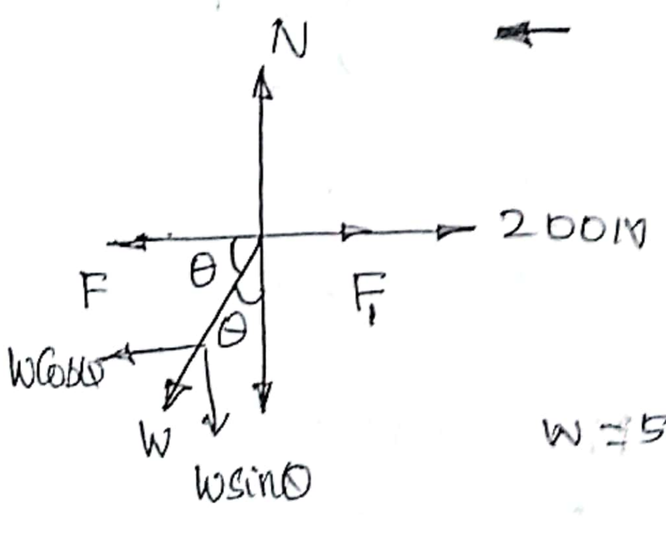
Solution:



N - Normal Reaction

Downward motion:

FBD



W = 500N

$\Sigma F_x = 0$

$-500\cos\theta + F_1 + 200 = 0$

$F_1 = \mu N$

$-500\cos\theta + \mu N + 200 = 0$

$\cancel{+}500\cos\theta = +\mu N \cancel{+}200$

$500\cos\theta = \mu N + 200$ —— (1)

$\Sigma F_y = 0$

$N - W\sin\theta = 0$

$N - 500\sin\theta = 0$

$N = 500\sin\theta$ —— (2)

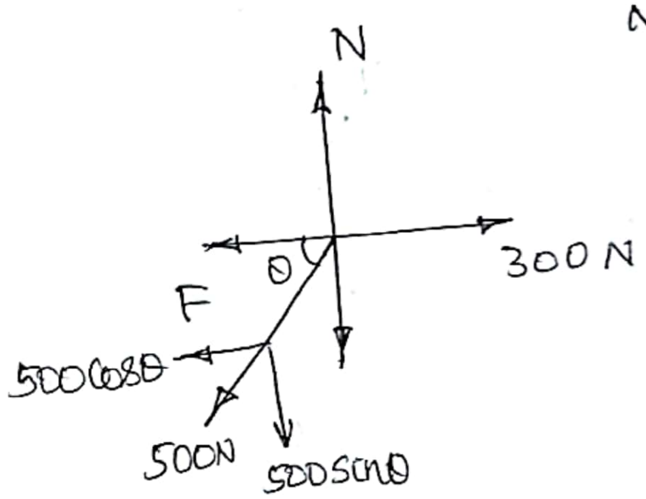
Sub in (1)

$500\cos\theta = 500\mu\sin\theta + 200$ —— (3)

<u>Upward Motion:</u>

<u>FBD</u>

N - Romal Reaction



F = μN

$\Sigma F_x = 0$

$-500\cos\theta - F + 300 = 0$

$500\cos\theta + F = 300$ —— (4)

$\Sigma F_y = 0$

$N - 500\sin\theta = 0$

$N = 500\sin\theta$

Sub in (4)

$500\cos\theta + \mu 500\sin\theta = 300$ —— (5)

(3) $\Rightarrow$ $500\cos\theta - 500\sin\theta\,\mu = 200$

(3) + (5) $1000\cos\theta = 500$

$\cos\theta = \frac{500}{100} = \frac{1}{2}$

$\theta = \cos^{-1}(1/2) = \underline{60^\circ}$

Sub. in (5),

$$500\cos\theta + 500\mu\sin\theta = 300$$

$$500\cos60^\circ + 500\mu\sin60^\circ = 300$$

$$433\mu + 250 = 300$$

$$433\mu = 300 - 250$$

$$= 50$$

$$\mu = \frac{50}{433} = \underline{\underline{0.115}}$$

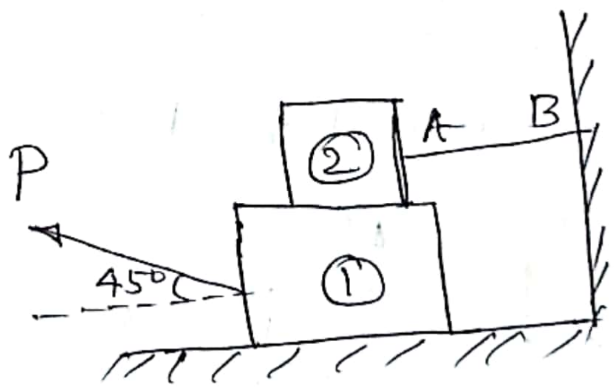
i) Co-efficient of friction, $\mu = 0.115$

ii) $\theta = 60^\circ$ (w.r.to X-X axis)

or

$\theta = 30^\circ$ (w.r.to Y-Y axis)

4. Block (2) rests on block (1) and is attached by a horizontal rope AB to the wall as shown in fig. What force P is necessary to cause motion of block (1) to impend? The coefficient of friction between the blocks is 1/4 and between the floor and Block (1) is 1/3. Mass of blocks (1) and (2) are 14 kg and 9 kg respectively (Dec. 2018)

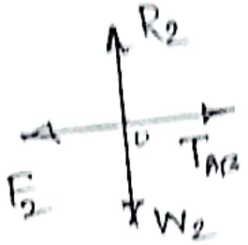


Solution:

Block (2):

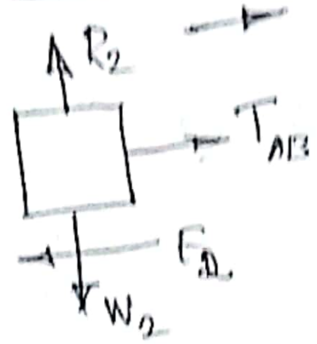
$\mu_2 = 0.25 = \frac{1}{4}$

$\mu_1 = \frac{1}{3} = 0.33$



(OR)

FBD



$W_1 = mg$
$= 14(9.81)$
$= 137.34N$

$W_2 = m_2 g$
$= 9 \times 9.81$
$= 88.29N$

Let, T_{AB} = Tension in the rope AB,

$\Sigma F_x = 0$

$-F_2 + T_{AB} = 0$

$F_2 = T_{AB}$ —— (1)

$F_2 = \mu_2 R_2$
$= 0.25 \times 88.29$
$= 22.072N$

$\Sigma F_y = 0$

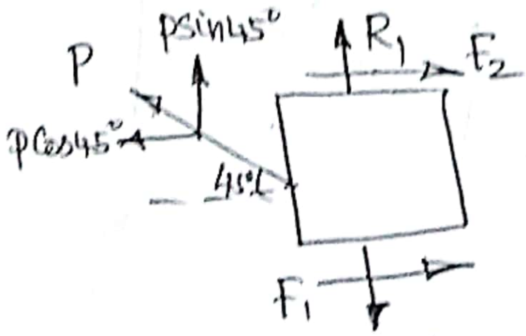
$R_2 - W_2 = 0$

$R_2 = W_2$
$= 88.29N$

sub F_2 in (1),

$\therefore T_{AB} = 22.072N$.

Block (1): FBD



$W = W_1 + W_2$
$= 137.34 + 88.29$
$= 225.63N$

$W(W_1 + W_2) = 225.63N$

F_1 and F_2 act towards Right by the application of force 'P'.

$\Sigma F_x = 0$

$-P\cos 45^\circ + F_1 + F_2 = 0$

$P\cos 45^\circ = F_1 + F_2$

$P\cos 45^\circ = 0.33R_1 + 22.072$ — (2)

$F_2 = \mu_2 R_2$
$= 0.25 \times 88.29$
$= 22.072N$

$F_1 = \mu_1 R_1$
$= 0.3R_1$

$\Sigma F_y = 0$

$R_1 + P\sin 45^\circ - W = 0$

$P\sin 45^\circ + R_1 - 225.63 = 0$

$P\sin 45^\circ = 225.63 - R_1$ —— (3)

$\frac{(3)}{(2)} \Rightarrow \frac{\cancel{P}\sin 45^\circ}{\cancel{P}\cos 45^\circ} = \frac{225.63 - R_1}{0.33R_1 + 22.072}$ | $\tan 45^\circ = 1$

$\tan 45^\circ = \frac{225.63 - R_1}{0.33R_1 + 22.072}$

$1 = \frac{225.63 - R_1}{0.33R_1 + 22.072}$

$0.33R_1 + 22.072 = 225.63 - R_1$

$0.33R_1 + R_1 = 225.63 - 22.072$

$R_1(0.33 + 1) = 203.558$

$R_1(1.33) = 203.558$

$R_1 = \frac{203.558}{1.33}$

$R_1 = 153.05N$

Sub. in (3),

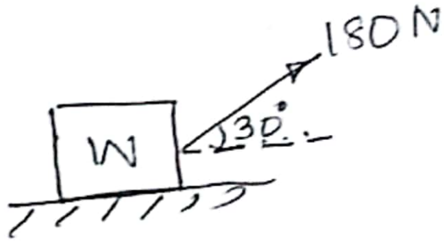
$P\sin 45^\circ = 225.63 - 153.05$

$P = \frac{72.63}{\sin 45^\circ} = 102.729N$

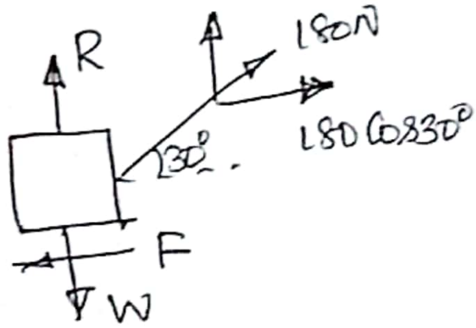
5. A body, resting on a rough horizontal plane required a pull of 180 N inclined at 30° to the plane just to move it. It was found that a push of 220 N inclined at 30° to the plane just moved the body. Determine the weight of the body and the co-efficient of friction (May 2017)

Solution:

Case 1: (pull)



FBD



$\Sigma F_x = 0$

$-F + 180 \cos 30^\circ = 0$

$F = 180 \cos 30^\circ$

$= 155.88\ N$

$F = \mu R$

$F = \mu R$

$155.88 = \mu R$ —— (1)

$\Sigma F_y = 0$

$-W + R + 180 \sin 30^\circ = 0$

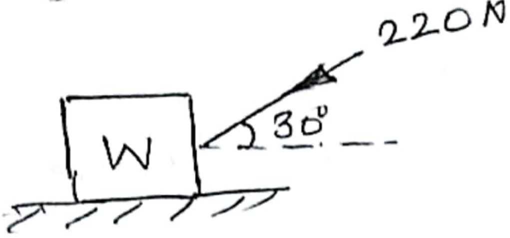
$-W + R + 90 = 0$

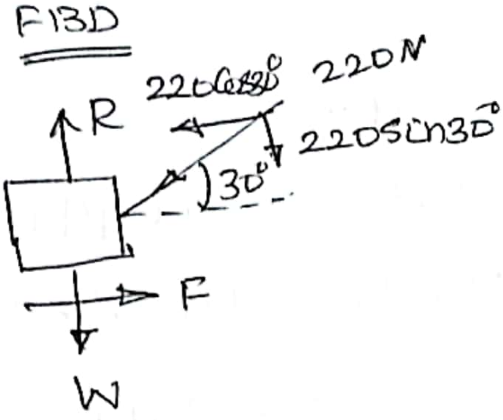
$R = W - 90$ —— (2)

Sub. in ①,

$$155.38 = \mu(W-90) \quad \text{③}$$

Case 2: (push)





$\Sigma F_x = 0$

$-220\cos30° + F = 0$

$F = 220\cos30°$

$= 190.52\ N$

$F = \mu R$

$$190.52 = \mu R \quad \text{④}$$

$\Sigma F_y = 0$

$R - W - 220\sin30° = 0$

$R = W + 220\sin30°$

$$R = W + 110 \quad \text{⑤}$$

Sub. in ④,

$$190.52 = \mu(W+110) \quad \text{⑥}$$

$$\frac{⑥}{③} \Rightarrow \frac{190.52}{155.98} = \frac{\cancel{\mu}(W+10)}{\cancel{\mu}(W-90)}$$

$$1.221 = \frac{W+10}{W-90}$$

$$1.221W - 109.9 = W + 110$$

$$1.221W - W = 110 + 109.9$$

$$W(1.22 - 1) = 219.9$$

$$0.22W = 219.9$$

$$W = \frac{219.9}{0.22} = 999.54\,N$$

$$= \cancel{545}\,N$$

Sub. in 6,

$$190.52 = \mu(999.54 + 110)$$

$$190.52 = 1109.54\mu$$

$$\mu = \frac{190.52}{1109.54}$$

$$= 0.17$$

i) weight of the body $= 999.54\,N$

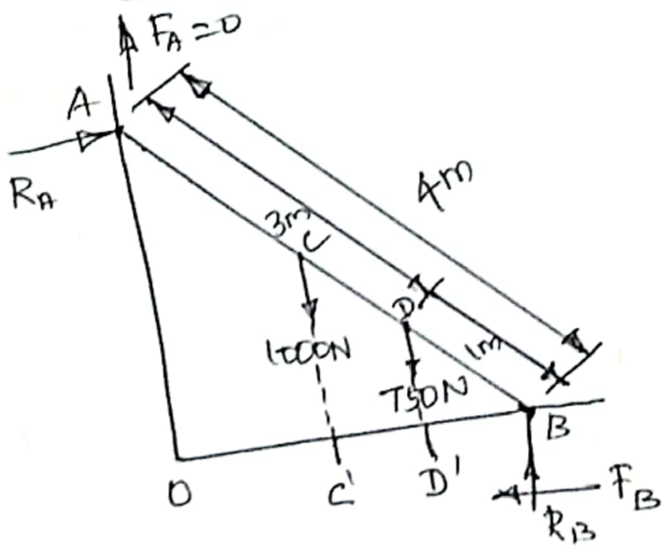
ii) Co-efficient of the friction, $\mu = 0.17$

Ladder

1. A uniform ladder of weight 1000N and of length 4 m rests on a horizontal ground and leans against a smooth vertical wall. The ladder makes an angle of 60° with horizontal. When a man of weight 750 N stands on the ladder at distance of 3 m from the top of the ladder the ladder is at the point of sliding. Determine the co-efficient of friction between the ladder and surface (May 2019)

Solution:

FBD



$\Sigma F_x = 0$

$R_A - F_B = 0$

$R_A = F_B$ $(F_B = \mu R_B)$

$= \mu R_B$

$R_A = \mu R_B$ —— (1)

$\Sigma F_y = 0$

$R_B + F_A - 1000 - 750 = 0$

$R_B + 0 - 1000 - 750 = 0$

$R_B = 1750N$

Sub. in (1),

$R_A = \mu R_B$

$R_A = 1750\mu$ ——— (2)

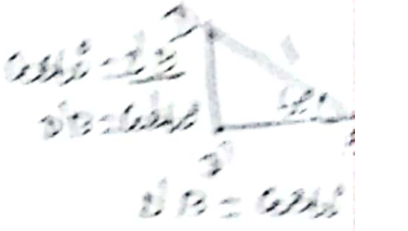
$\Sigma M_B = 0$

$-(R_A \times 4\sin 60^\circ) + (1000 \times 2\cos 60^\circ) + (750 \times \cos 60^\circ) = 0$

$-R_A \times 3.464 + 1000 + 375 = 0$

$+R_A \times 3.464 = +1375$

$R_A = \dfrac{1375}{3.464} = \underline{\underline{396.94\ N}}$



Sub. in (2),

$R_A = 1750\mu$

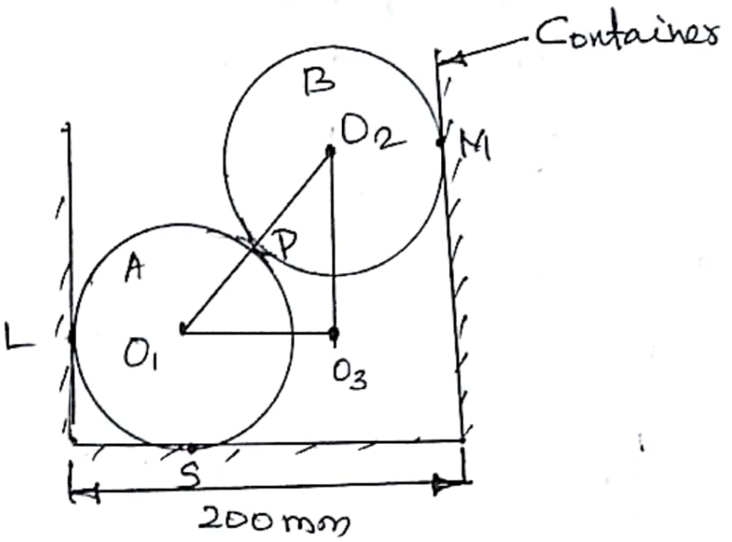
$396.94 = 1750\mu$

$\mu = \dfrac{396.94}{1750}$

$= \underline{\underline{0.226}}$

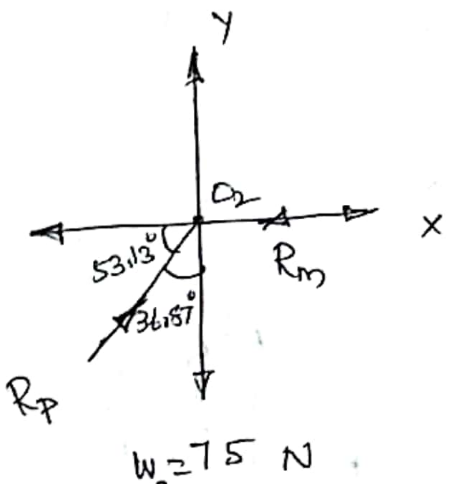
The co-efficient of friction between ladder and floor surface $\underline{\underline{\mu = 0.226}}$

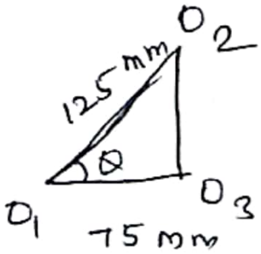
2) Two sphere A and B weighing 100 N and 75 N respectively and with corresponding radii 75 mm and 50 mm are placed in a container as shown in fig. Determine the support reactions (May 2019)



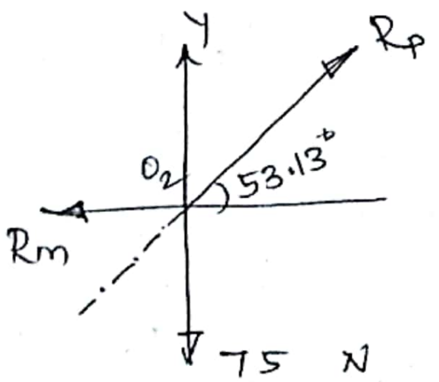
Solution:

Roller B: FBD





Applying principle of transmissibility



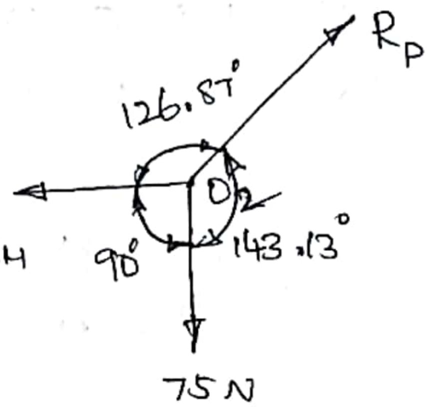
$$\cos\theta = \frac{75}{125}$$

$$\theta = \cos^{-1}\left(\frac{75}{125}\right)$$

$$= 53.13°$$

Applying Lami's theorem

FBD



$$\frac{R_H}{\sin 143.13°} = \frac{R_P}{\sin 90°} = \frac{75}{\sin 126.87°}$$

$$\frac{R_H}{\sin 143.13°} = \frac{75}{\sin 126.87°}$$

$$R_H = \frac{75 \times \sin 143.13°}{\sin 126.87°}$$

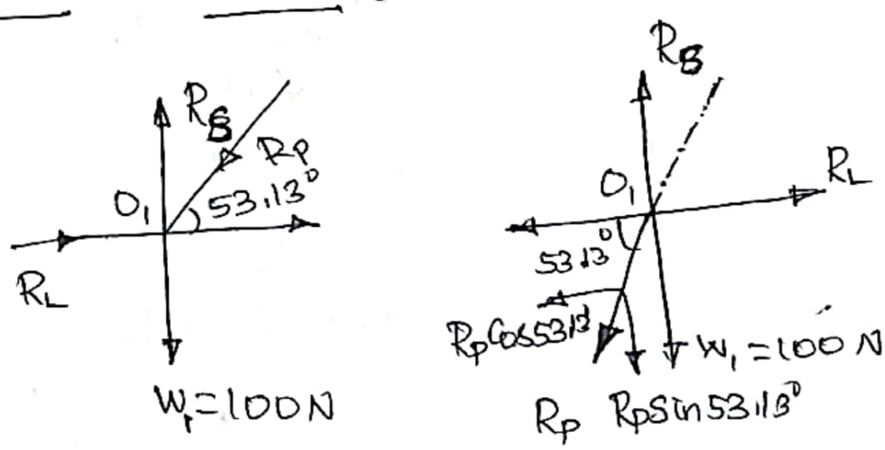
$$= \underline{\underline{56.25\,N}}$$

$$\frac{R_P}{\sin 90°} = \frac{75}{\sin 126.87°}$$

$$R_P = \frac{75 \times \sin 90°}{\sin 126.87°}$$

$$= \underline{\underline{93.75\,N}}$$

Roller A : FBD



$\Sigma F_y = 0$

$$R_S - R_P \sin 53.13° - 100 = 0$$

$$R_S = R_P \sin 53.13° + 100$$

$$= \underline{\underline{175\,N}}$$

$\Sigma F_x = 0$

$$R_L - R_P \cos 53.13° = 0$$

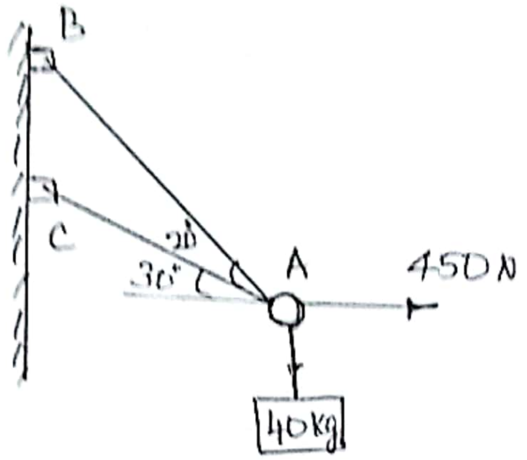
$$R_L = R_P \cos 53.13°$$

$$= 93.75 \cos 53.13° = \underline{56.25\,N}$$

$$R_L = \underline{\underline{56.25\,N}}$$

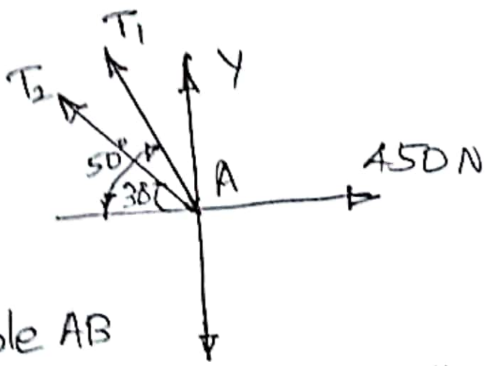
Scanned with CamScanner

* Determine the tension in cables AB and AC required to hold the 40 kg crate shown in fig. (Sep.2020)



Solution:

FBD



Let,

T_1 = Tension in cable AB

T_2 = Tension in Cable AC

$W = 40 \times 9.81$
$= 392.4 N$

$W = mg$
$= 40 \times 9.81$
$= 392.4 N$

$\Sigma F_x = 0$

$\Sigma F_x = 450 - T_1 \cos 50^\circ - T_2 \cos 30^\circ = 0$

$\Sigma F_x = 450 - 0.643 T_1 - 0.877 T_2 = 0$

$0.643 T_1 + 0.877 T_2 = 450$ —— (1)

$\Sigma F_y = 0$

$0.77 T_1 + 0.5 T_2 - 392.4 = 0$

$0.77 T_1 + 0.5 T_2 = 392.4$ —— (2)

(1) $\times 0.77 \Rightarrow 0.495 T_1 + 0.669 T_2 = 346.50$ —— (3)

(2) $\times 0.643 \Rightarrow 0.495 T_1 + 0.321 T_2 = 252.31$ —— (4)

(3) − (4) $\Rightarrow$ $\cancel{0.495 T_1} + 0.669 T_2 = 346.50$
$\cancel{0.495 T_1} + 0.321 T_2 = 252.31$

$T_2 = 270.34 N$

Sub. in (1),

$$0.643T_1 + 0.87(270.34) = 450$$

$$\underline{T_1 = 334.06 N}$$